Topics of today's class

Last class:

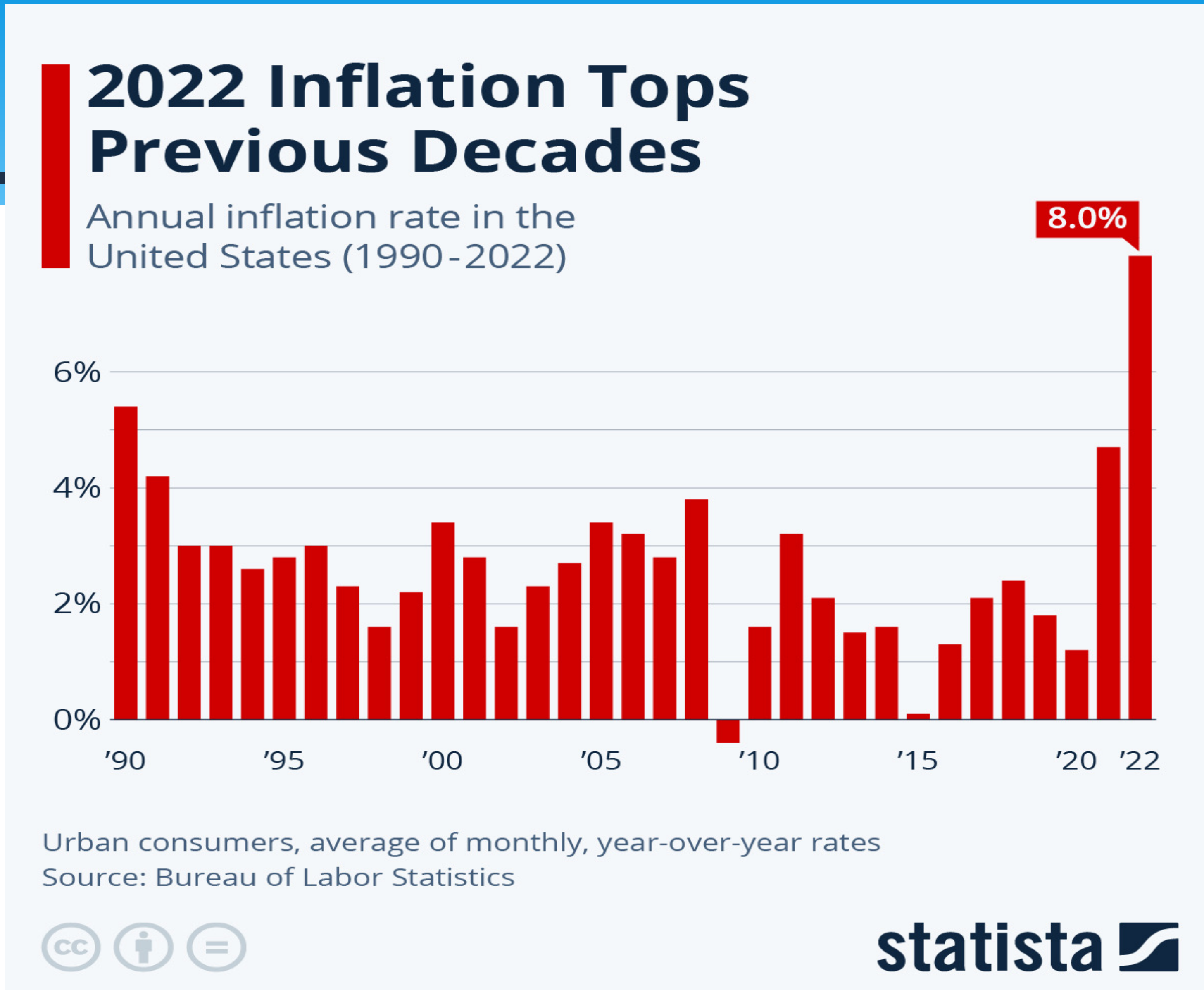
- Chapter 2
  - Measuring GDP using three approaches
  - Problems in measuring GDP

Today's class:

- Chapter 2
  - Nominal and Real GDP and Price Indices
  - Savings, wealth and capital
  - Labor market measurement

Chapter 2

Measurement


2022 Inflation Tops Previous Decades
Annual inflation rate in the United States (1990-2022)
8.0%
6%
4%
2%
0%
'90
'95
'00
'05
'10
'15
'20
'22
Urban consumers, average of monthly, year-over-year rates
Source: Bureau of Labor Statistics
statista

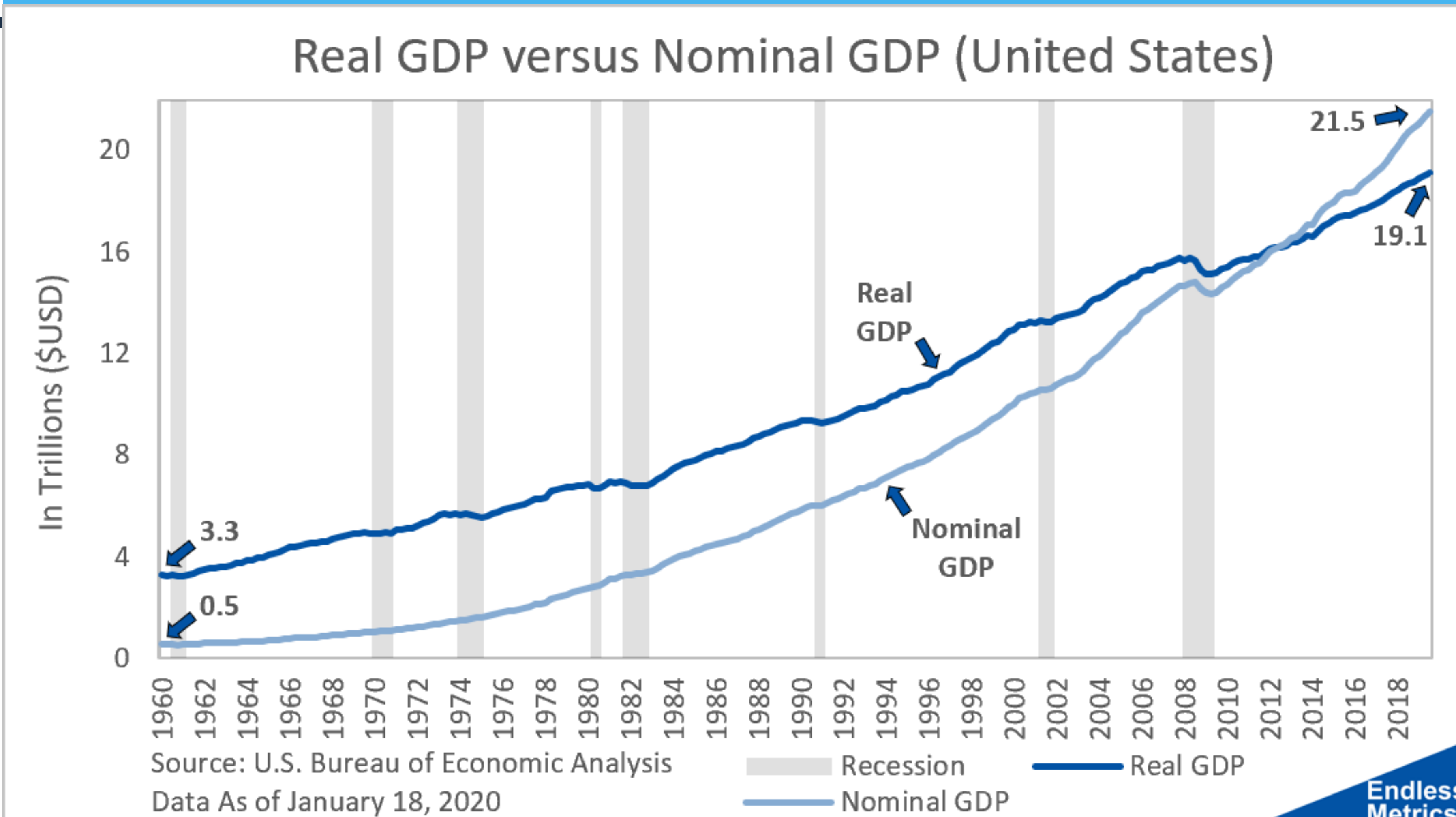
Nominal, Real GDP and Price Indices

- Nominal GDP changes between year *t* and year *t + 1* because:
  - different amounts of goods and services are produced between the two years.
  - such goods and services are sold at different prices in the two years.
- However, standards of living are determined by the quantities of goods and services produced and consumed by households, not by the nominal value they have in the market. If we are interested in quantities, why don't we just sum quantities and forget about prices altogether?
  - You cannot sum 2 apples, 1 car and 3 plane tickets: what's the result? We need prices to convert all goods and services in $ and aggregate them together

Nominal, Real GDP and Price Indices

- Price Index: Weighted average of a set of observed prices that gives a measure of the *price level*.
- Price indices allow us to measure the *inflation rate* – the rate of change in the price level.
- A measure of the inflation rate allows us to determine how much of an increase in GDP is *nominal* and how much is *real*.

Figure 2.1
Nominal GDP and Chain-Weighted Real GDP for the Period 1947–2009.



Measures of the Price Level: CPI

There are two commonly used measures of the price level:

- **Consumer Price Index (CPI)**
  - only include goods and services that are purchased by consumers: “measure of standard of livings”.
  - **fixed basket index**: define the typical quantities of goods and services that are purchased by consumers
  - compute the average price of this basket of goods each year.

Current year CPI =
Cost of base year quantities at current year prices/
Cost of base year quantities at base year prices x 100

Measures of the Price Level: GDP Deflator

- **Implicit GDP Price Deflator**
  - **time-varying basket index**

    Implicit GDP Price Deflator
    = Nominal GDP / Real GDP x 100
    = Cost of current year quantities at current year prices
    / Cost of current year quantities at base year prices x 100.

- The key difference between **CPI** and **GDP Deflator** is that in one the weights do not change but in the other the weights change over time.

Table 2.10
Data for Real GDP Example

Table 2.10 Data for Real GDP Example

	Apples	Oranges
Quantity in Year 1	$Q_1^a = 50$	$Q_1^o = 100$
Price in Year 1	$P_1^a = \$1.00$	$P_1^o = \$0.80$
Quantity in Year 2	$Q_2^a = 80$	$Q_2^o = 120$
Price in Year 2	$P_2^a = \$1.25$	$P_2^o = \$1.60$

GDP Deflator:
Year 1: (50*1+100*0.8) / (50*1+100*0.8) * 100 = 100
Year 2: (80*1.25+120*1.6) / (80*1+120*0.8) * 100 = 165.9

Table 2.11
Implicit GDP Price Deflators, Example

Table 2.11 Implicit GDP Price Deflators, Example

	Year 1	Year 2	% Increase
Year 1 = base year	100	165.9	65.9
Year 2 = base year	58.4	100	71.2
Chain-weighting	100	168.5	68.5

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CPI Deflator:

Year 1: (50*1+100*0.8) / (50*1+100*0.8) * 100 = 100

Year 2: (50*1.25+100*1.6) / (50*1+100*0.8) * 100 = 171.2

Figure 2.2

Inflation Rate Calculated from the CPI and from the Implicit GDP Price Deflator

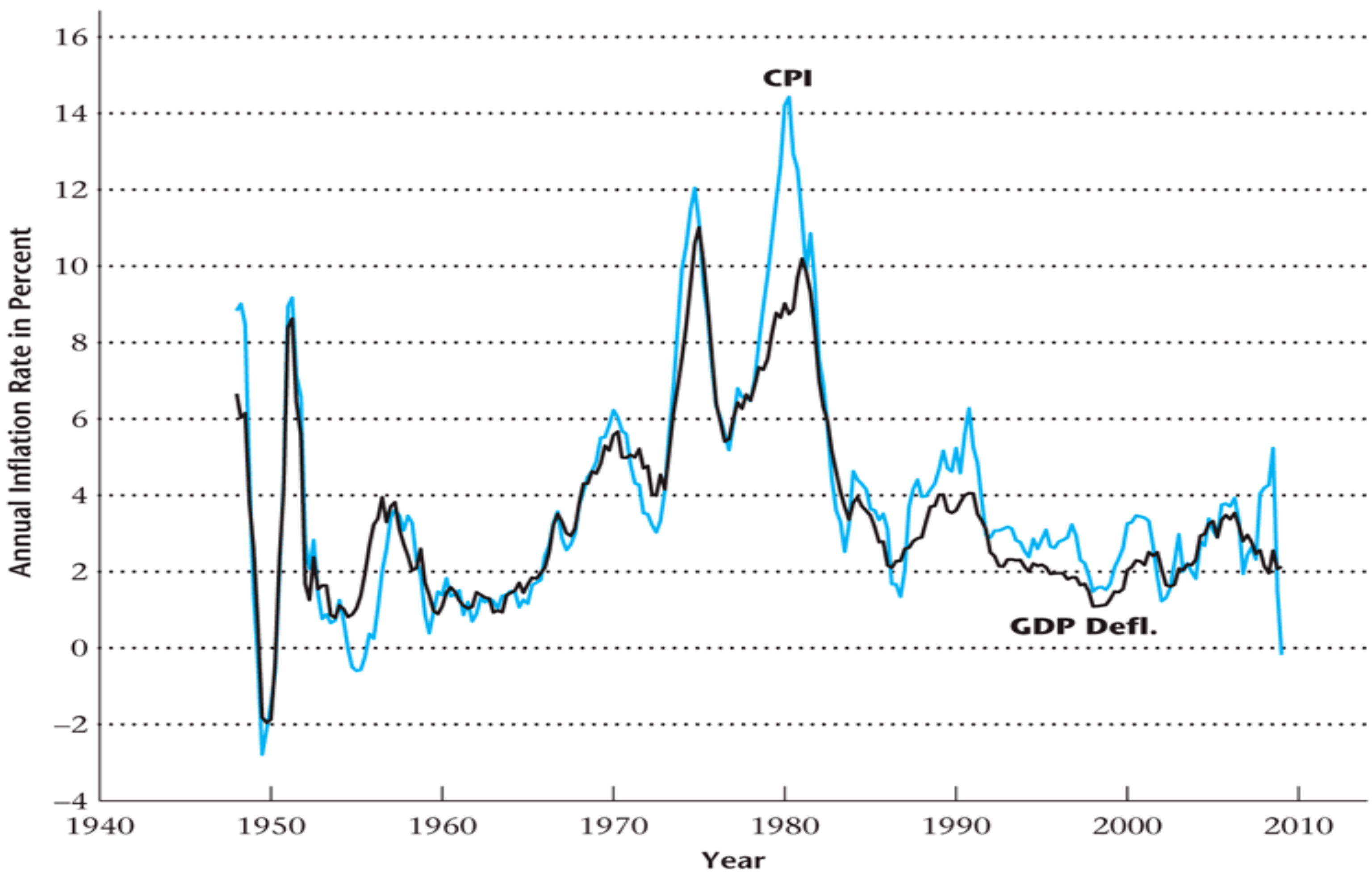
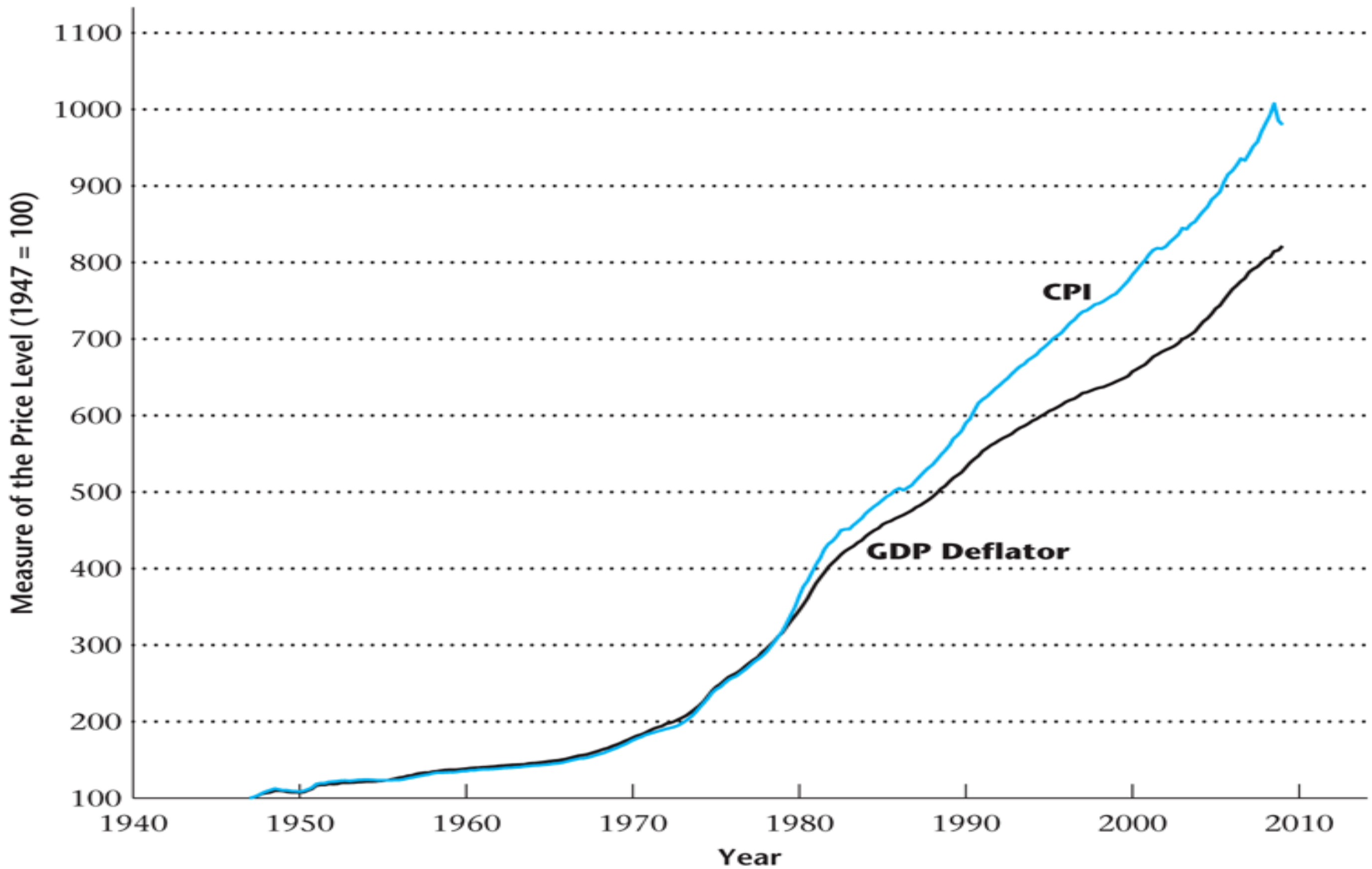


Figure 2.3
The Price Level as Measured by the CPI and the Implicit GDP Price Deflator, 1947–2009



CPI v.s. GDP Deflator

- CPI is more volatile.
- CPI is upward biased: substitution bias.
  - oil price shock: price of oil skyrocketed in 1974-75.
  - households reduce the demand for gasoline and cars and increase their demand for public transportation.
  - the basket of goods includes much less gasoline in 1976.
  - then the CPI using the 1974 quantities will overstate inflation.

BUT:

- CPI is better measures of the cost of living in the US.
- GDP price deflator based on all goods produced in the economy, even investment goods (such as trucks, planes, oil pipes) which are not of interest to households.

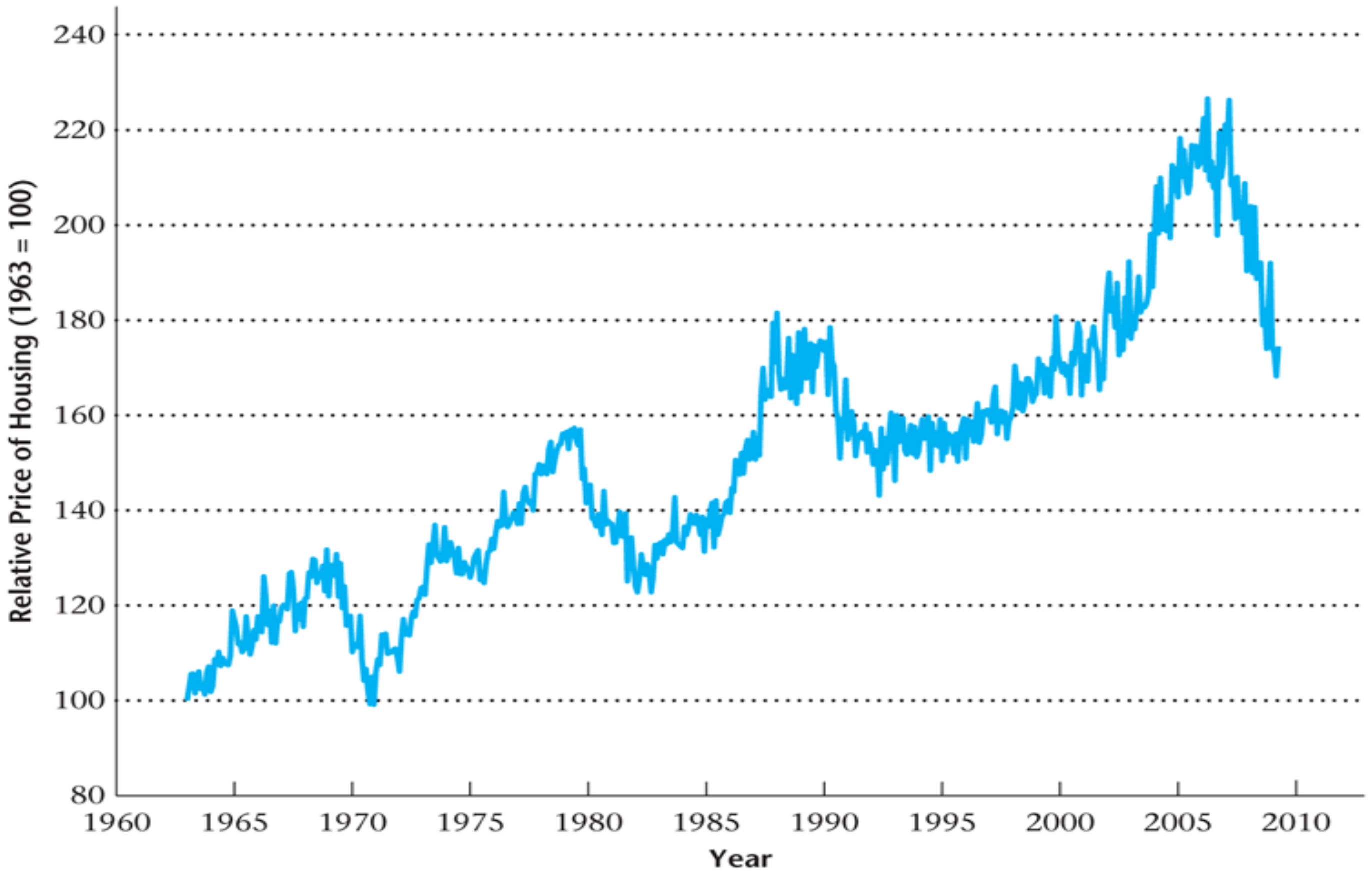
Problems in Measuring Real GDP and the Price Level

- The relative prices of goods change over time – a problem for CPI measurement.
- The quality of goods and services changes over time.
- New goods and services are introduced, and some goods and services become obsolete.

House Prices and GDP Measurement

- As a result of incentive problems in the mortgage market, the relative price of houses may have exceeded its true economic value prior to 2006.
- The relative price of housing fell dramatically beginning in 2006.
- This overvaluation of new houses may have exaggerated the value added in GDP accounted for by housing construction – an upward bias of perhaps 1.2%.

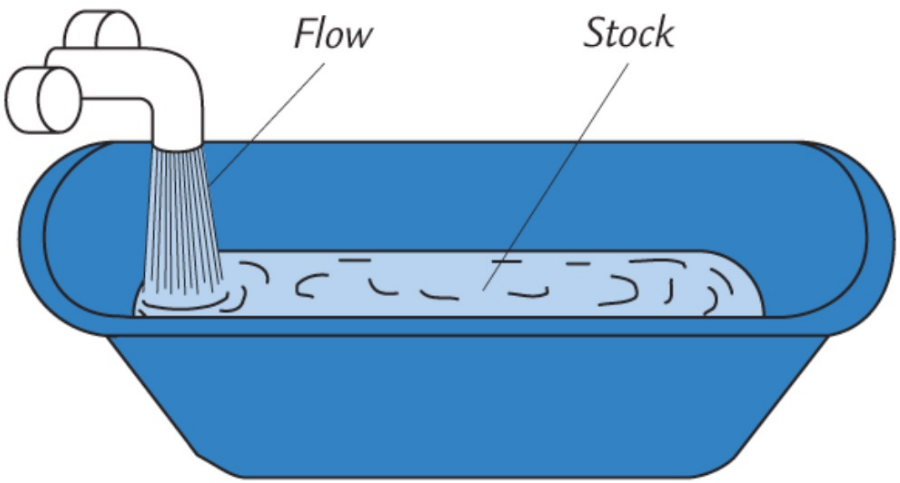
Figure 2.4
The Relative Price of Housing in the United States



Savings and Investment

- A stock is a quantity measured at a point in time.
  E.g., "The U.S. capital stock was $10 trillion on January 1, 2012."

- A flow is a quantity measured per unit of time.
  E.g., "U.S. investment was $2 trillion during 2012."



Savings and Investment

stock	flow
a person's wealth	a person's annual saving
# of people with college degrees	# of new college graduates this year
the govt debt	the govt budget deficit

Savings and Investment

- the balance on your credit card statement
  Stock
- Population of the U.S.
  Stock
- the size of your compact disc collection
  Stock
- the number of workers who lost their jobs last week
  Flow

Savings and Investment

- ***Y^d***: private disposable income.
- ***Y*** : GDP=all income received by agents contributing to production.
- ***T***: Taxes

It holds:

$$Y^d = Y - T.$$

Savings and Investment

- S^p: private sector saving

$$S^p = Y^d - C$$
$$= Y - T - C$$

Savings and Investment

- S^g: government saving
- G: government expenditures
- D: government deficit

It holds:

$$S^g = T - G$$

$$D = -S^g = G - T$$

Savings and Investment

- $\boldsymbol{S}$: national saving

$$\boldsymbol{S = S^p + S^g}$$
$$\boldsymbol{= (Y - T - C) + (T - G)}$$
$$\boldsymbol{= Y - C - G}$$

Savings and Investment

- ***GDP***=total spending on all final goods and services production in the economy

$$\boldsymbol{Y = C + I + G + NX}$$

Combined with expression on national saving:

$$\begin{aligned} \boldsymbol{S} &= \boldsymbol{Y - C - G} \\ &= \boldsymbol{C + I + G + NX - C - G} \\ &= \boldsymbol{I + NX} \end{aligned}$$

Savings and Investment

Re-writing:

$$S^p = I + D + NX$$

Private sector saving is used to finance:

1. private sector investment
2. government deficit
3. current account deficit

Savings and Investment

$$S^p = I + D + NX$$

Be careful this is just an accounting equality: an increase in ***D*** does not necessarily imply a decrease in ***I***

$$\Uparrow D \Rightarrow \Uparrow Y \Rightarrow \Uparrow S^p$$

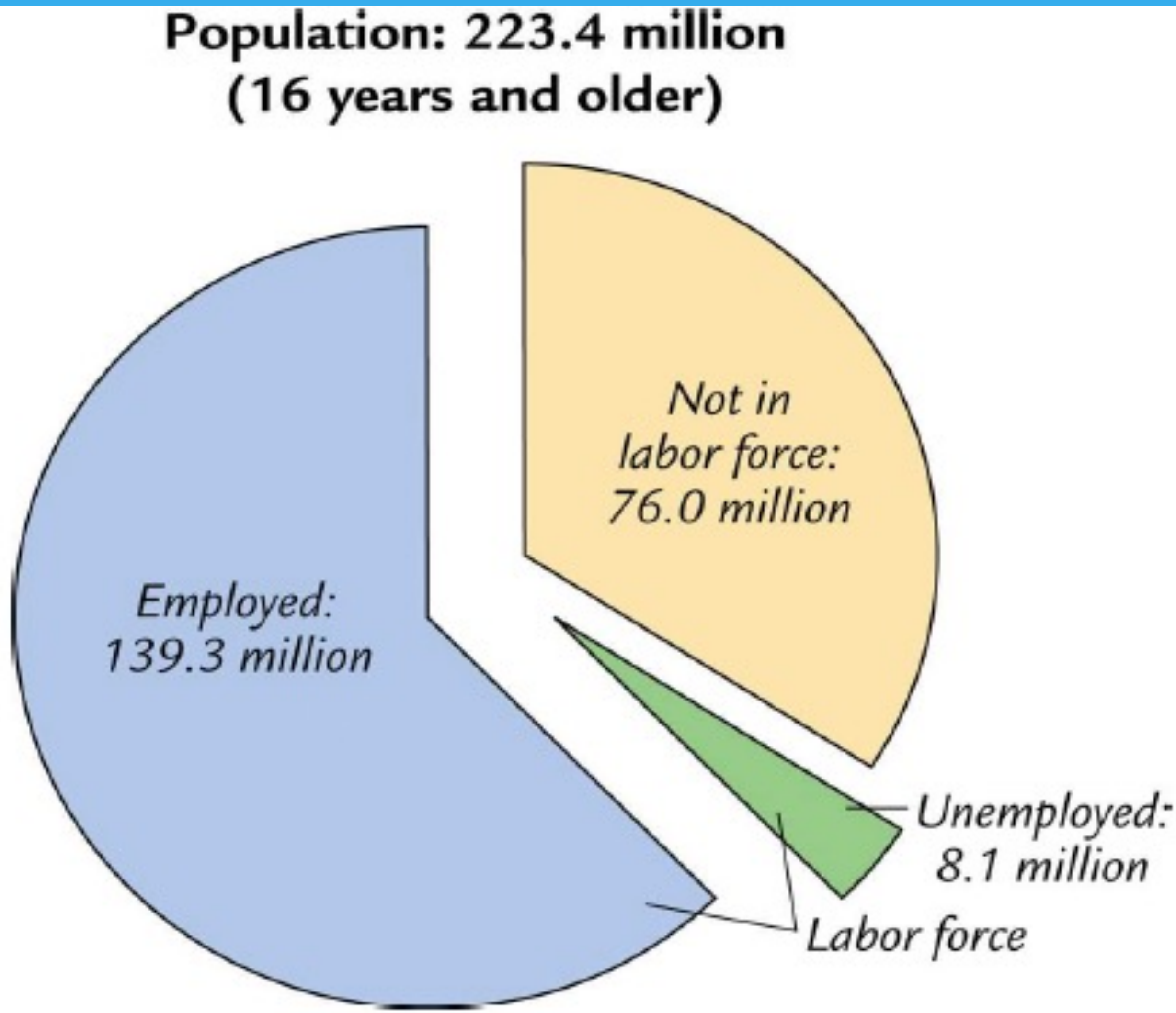
Labor Market Measurement

- Adult population / potential workers: Everyone in the general population except:
  1. Children under 16 years
  2. Active military personnel
  3. Institutionalized persons

Labor Market Measurement

- BLS Categories
  - Employed
  - Unemployed: unemployed during the past week but actively searched for work during the last four weeks
  - Not in the Labor Force

Labor Market Measurement



Labor Market Measurement

- ***employed***: working as paid employee, working in their own business, or working as unpaid workers in a family member's business. Also include those who are absent due to pregnancy, illness, vacation, etc.
- ***unemployed***: not employed but looking for a job in the previous 4 weeks. Also include those waiting to be recalled to a job from which they had been laid off.
- ***labor force***: the amount of labor available for producing goods and services; all employed plus unemployed persons
- ***not in the labor force***: Neither employed, nor looking for work. Too old, full-time student, disabled, too rich

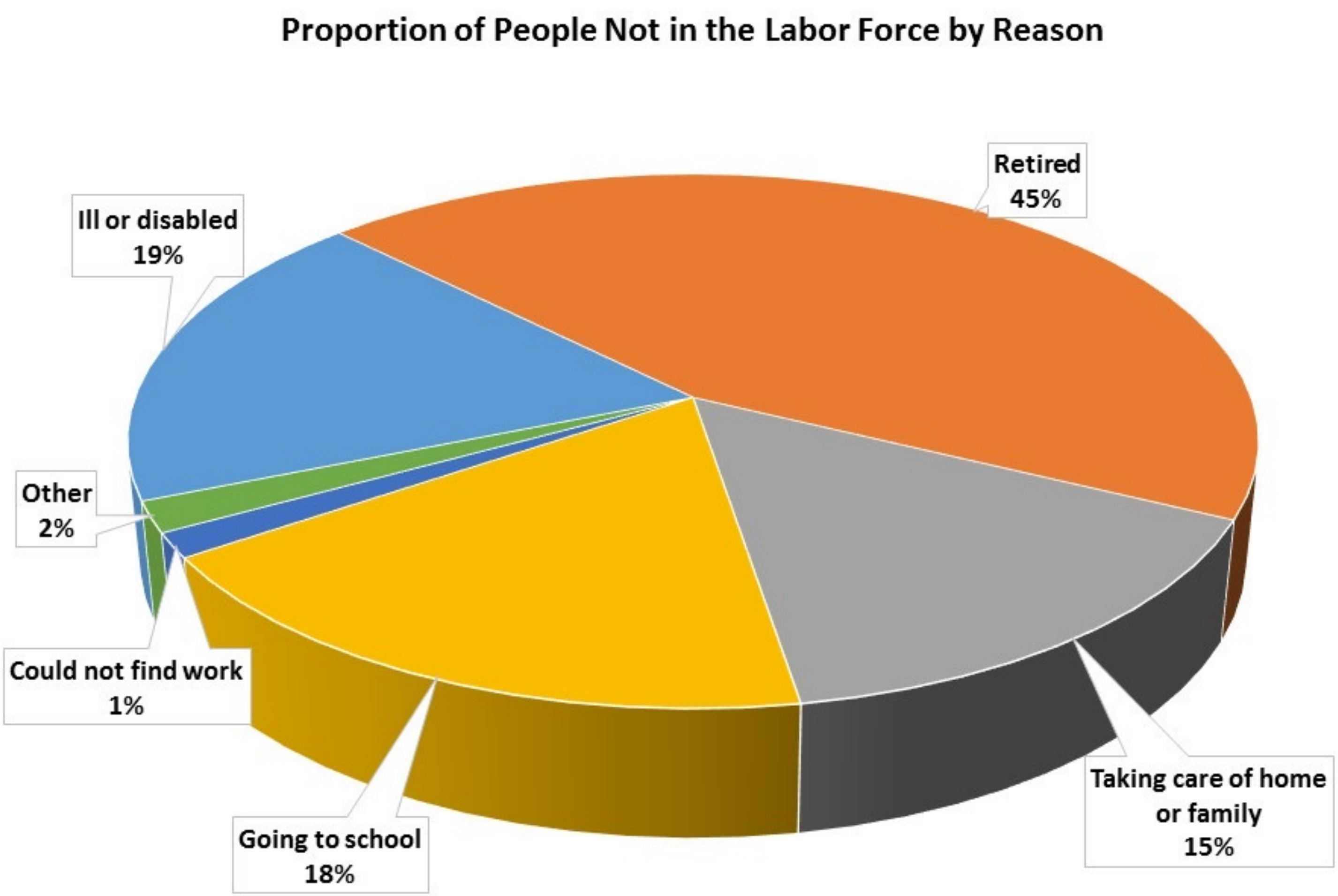
Examples

- A college student working part-time. *employed*
- A female employee on maternity leave. *employed*
- John is a university teacher, and in 2011 he won the powerball lottery of 146 million dollars. Then he quit his job, bought a beachfront villa in Hawaii and has been living there ever since. *not in the labor force: too rich to work*
- Mr Badluck lost his job last year. He is hunting for a new job now. *unemployed*
- Ms Badluck lost her job last year. She has been looking for a new job in the last 6 months, but could not find one. She gave up and decided to live with her parents for one year. *not in the labor force: discouraged worker*

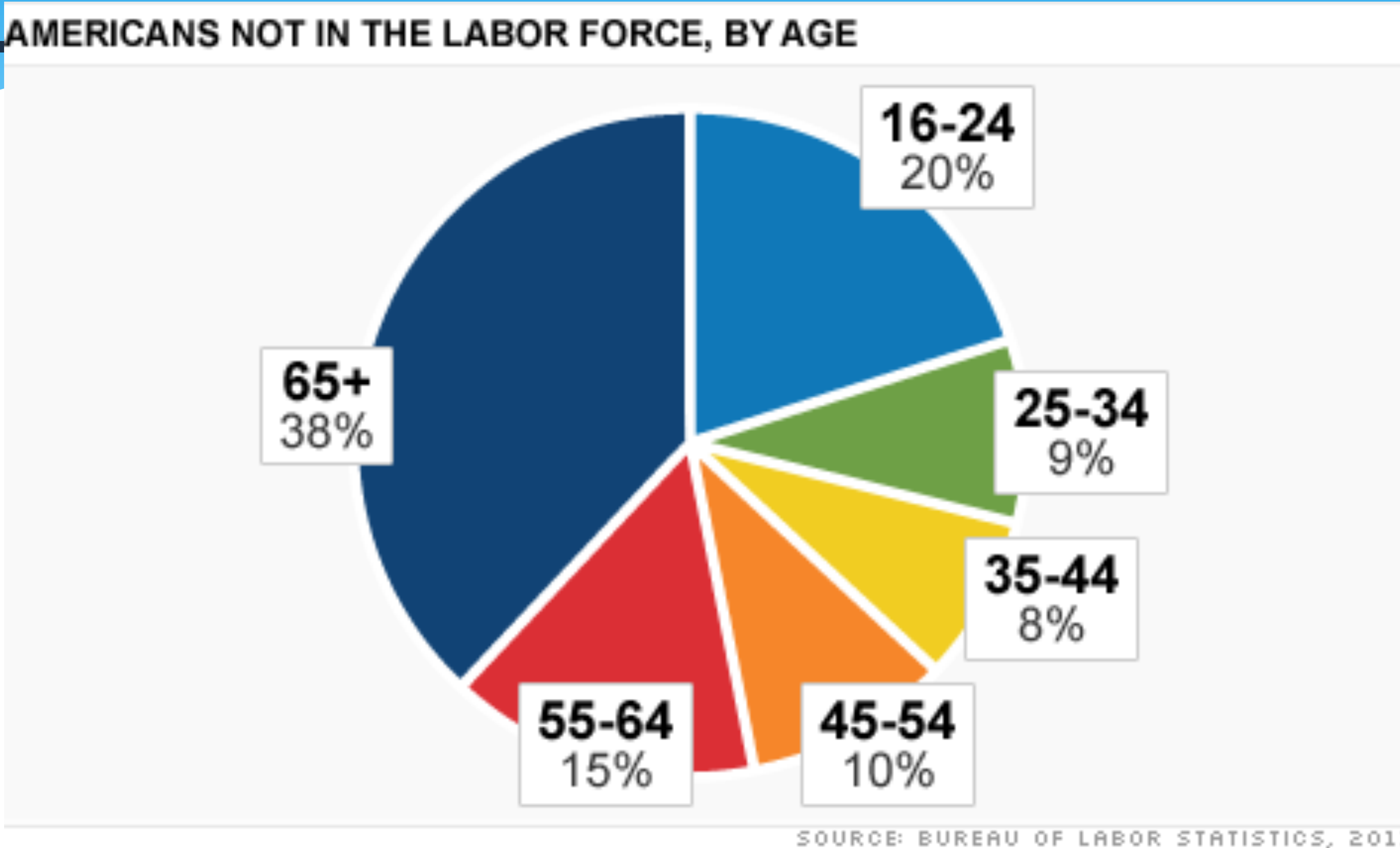
Labor Market Measurement

- Labor Force = Employed + Unemployed
- The Unemployment Rate =
  Number unemployed / Labor force
- Participation Rate =
  Labor force / Total working age population

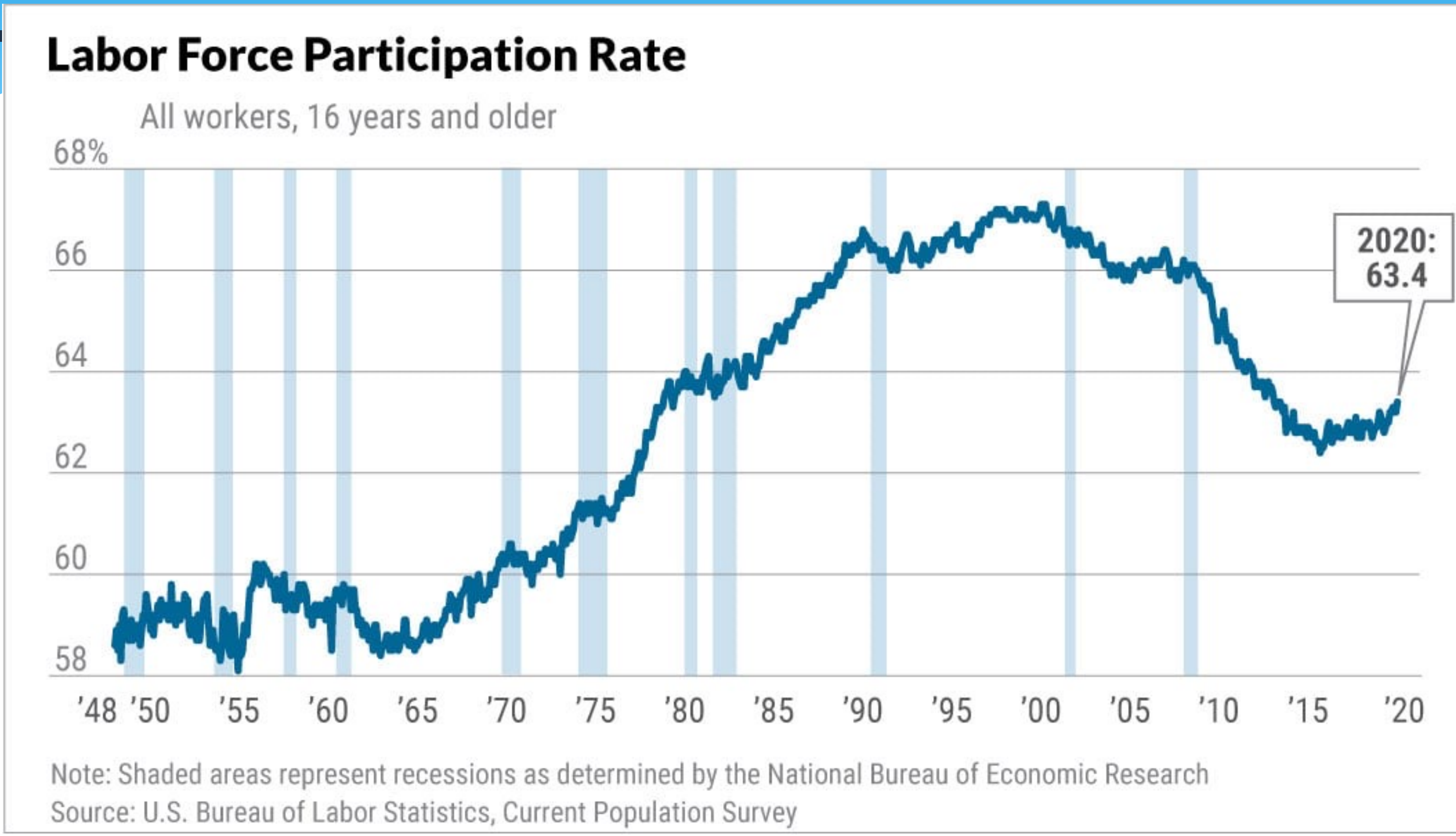
Labor Market Measurement



Labor Market Measurement



Trends in labor force participation



NOW YOU TRY

- U.S. adult population by group, May 2009
  - Number employed = 140.57 million
  - Number unemployed = 14.51 million
  - Adult population = 235.45 million

- Use the above data to calculate
  - the labor force
  - the number of people not in the labor force
  - the labor force participation rate
  - the unemployment rate

Answers

Data: $E = 140.57$, $U = 14.51$, $POP = 235.45$

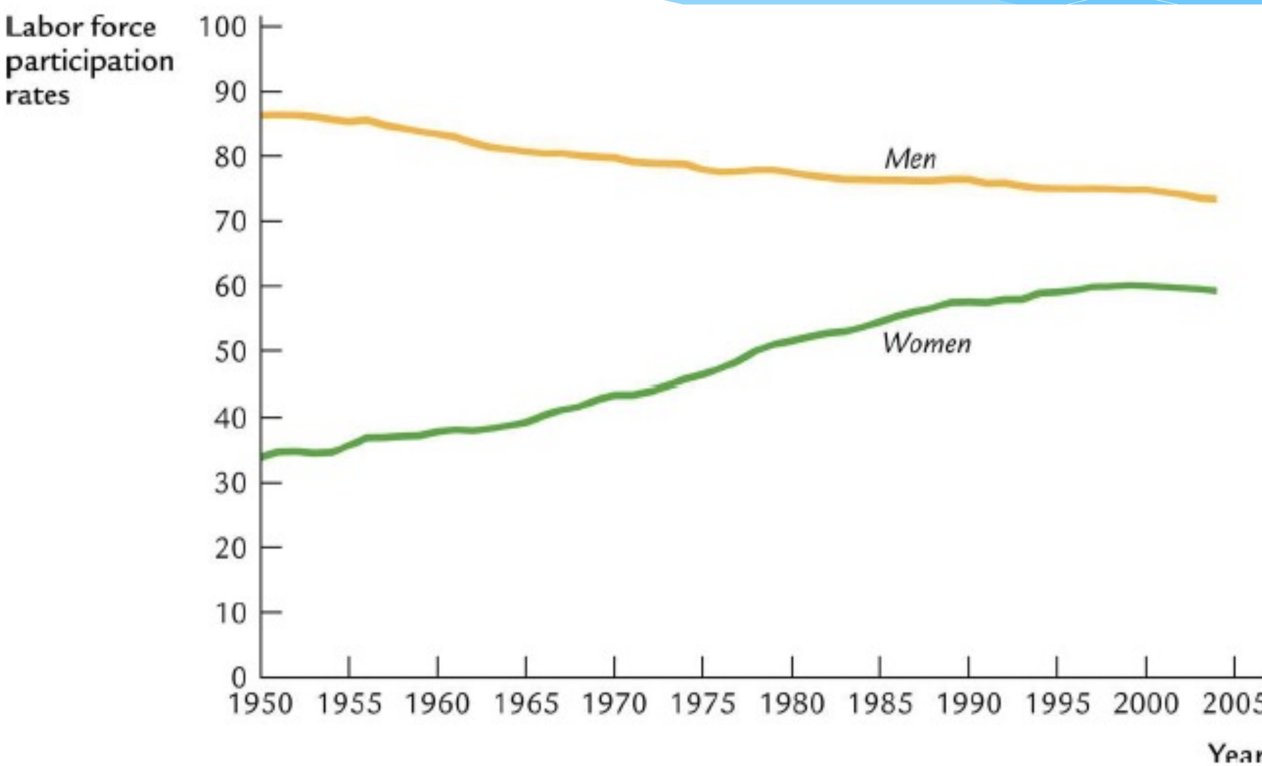
- labor force:
  $L = E + U = 140.57 + 14.51 = 155.08$

- not in labor force:
  $NILF = POP - L = 235.45 - 155.08 = 80.37$

- unemployment rate
  $U/L \times 100\% = 14.51/155.08 \times 100\% = 9.4\%$

- labor force participation rate
  $L/\ POP \times 100\% = 155.08/\ 235.45 \times 100\% = 65.9\%$

Case study: gender wage gap, labor force participation and Obesity



Case study: gender wage gap, labor force participation and Obesity

Households	1960	1990
Married couples		
Hours worked (female)	10.7	22.2
Hours worked (male)	39.4	38.9
Single females		
Hours worked	22.4	24.7
Single males		
Hours worked	27.9	27.8

Households	1965	1995
Married couples		
Hours food prep. (female)	13.0	6.4
Hours food prep. (male)	1.2	1.7
Single females		
Hours food prep.	7.0	3.8
Single males		
Hours food prep.	2.1	2.1